

# Community Audio Broadcast to 500 Households — BRD

2026-09-21 · @Someone

*Weekly audio distribution from a church or masjid to ~500 households within a 2 km radius. Pilot-first, platform-ready.*

## Executive summary

Build a second, private audio channel that carries what the outdoor loudspeakers carry today — the weekly announcement and up to five adhan a day — so that if loudspeaker permission is curtailed the institution does not lose contact with its 500 households.

The decisive constraint is regulatory, not technical. A dedicated radio broadcast is not available to this institution: the Ministry of Information and Broadcasting's community radio guidelines list **religious bodies as ineligible** to hold a licence, and the 87.5–108 MHz band is otherwise open only to nanowatt short-range devices, which cannot cover 2 km. Every RF option that would reach 500 scattered homes either needs a licence the institution cannot obtain or is unlawful. The design therefore rides infrastructure that already exists in the households: home broadband, mobile data, and the ordinary telephone network.

### Recommended shape — one live stream, three household endpoints:

| Tier | Household endpoint                                       | One-time cost to household | Share of 500 houses (assumed) |
|------|----------------------------------------------------------|----------------------------|-------------------------------|
| T1   | Own smartphone, web link or app                          | ₹0                         | ~55% (275)                    |
| T2   | "Parish Box" — Wi-Fi speaker, always on, one volume knob | ₹285                       | ~30% (150)                    |
| T3   | Any phone rings and plays the live audio                 | ₹0                         | ~15% (75)                     |

No household pays more than **₹285** once, against the ₹300 ceiling. T3 needs no device purchase at all and covers feature-phone and elderly households, which is where a device-based design normally fails.

**Two product versions.** *Basic* is live only and costs about **₹1.46 lakh** to build. *Pro* adds replay, archive, an app, transcription and redundancy for about **₹3.48 lakh**. Both sit inside the ₹1–4 lakh envelope. Running cost is about **₹80,000/year** for Basic and **₹1.31 lakh/year**

for Pro — roughly **₹219 per household per year** over five years for Basic, covering five daily adhan and the weekly announcement.

**Migration.** The public address system keeps running unchanged throughout the pilot. The new channel is additive, proves itself against a working baseline, and is already in place on the day any restriction lands. That, not the hardware, is the point of the project.

All costs in this document are planning estimates in Indian rupees at September 2026 prices, for a 500-household pilot, and must be re-quoted before purchase.

## Background and problem statement

The institution already runs a working one-to-many audio channel, and it costs almost nothing per household. That is the baseline any replacement has to beat.

**How it works today.** A microphone and amplifier inside the church or masjid drive horn loudspeakers mounted on the building or a tower. Audio radiates outward and reaches roughly 500 households within about 2 km. Households need no device, no account, no enrolment and no action — sound simply arrives. Marginal cost per household is zero, and the operator's effort per broadcast is switching on an amplifier and speaking.

**What it carries.** Adhan five times a day, weekly announcements, prayer and service times, community notices, deaths and funeral timings, relief and emergency calls, festival and schedule changes.

**Why it is at risk.** Outdoor amplification is governed by the Noise Pollution (Regulation and Control) Rules, 2000 under the Environment (Protection) Act. Using a loudspeaker in a public place requires **written permission** from the designated authority, is barred between 10 pm and 6 am outside closed premises, and must keep the noise level at the boundary within 10 dB(A) above ambient or 75 dB(A), whichever is lower. Indian courts have repeatedly held that amplified sound is not itself a protected religious practice — in *Church of God (Full Gospel) in India v. K.K.R. Majestic Colony Welfare Association* the Supreme Court held that the right to profess and propagate religion is not absolute where it affects others' rest, and the Bombay High Court has held that loudspeaker use is not an essential part of any religion. Permissions are granted annually, at district discretion, and enforcement has tightened in several states.

**The gap this project closes.** The institution has no second channel. If permission is refused, narrowed to certain hours, or suspended during an enforcement drive, weekly contact with 500 households stops that week. There is no register of who was reached, no way to reach people who are indoors, at work, or outside the sound cone, and nothing to fall back on.

| Attribute                      | Loudspeaker PA (today)                  | What the replacement must match      |
|--------------------------------|-----------------------------------------|--------------------------------------|
| Household one-time cost        | ₹0                                      | ≤ ₹300                               |
| Household action per broadcast | None                                    | At most one tap or one answered call |
| Enrolment needed               | None                                    | One-time, assisted                   |
| Reach                          | ~500 homes within 2 km, outdoor-audible | 500 named households, indoors        |
| Operator effort                | Switch on, speak                        | Comparable; no technician present    |
| Regulatory exposure            | High and rising                         | None                                 |
| Delivery confirmation          | None                                    | Per-household, for the first time    |

## Regulatory context (India)

Four distinct regimes apply. Together they eliminate every broadcast-radio option and shape how the telephone and internet options must be built.

### 1. Noise and loudspeakers — the risk being hedged

The [Noise Pollution \(Regulation and Control\) Rules, 2000](#) require written permission from the designated authority for any loudspeaker or public address system in a public place, bar use between 10 pm and 6 am except in closed premises or a public emergency, and cap boundary noise at ambient + 10 dB(A) or 75 dB(A), whichever is lower. States may relax the night restriction for up to 15 days a year for festivals. Permission is discretionary and renewable, so it can narrow or lapse without a change in law. **Implication:** the project is insurance against a permission that may not be renewed, and the new channel must be independent of any loudspeaker permission.

### 2. Broadcast radio — closed to this applicant

The [Revised Policy Guidelines for Community Radio Stations \(13.02.2024\)](#) limit eligibility to not-for-profit community organisations and educational institutions with three or more years of community service, and clause 2(b) expressly lists **religious bodies** among those who may not hold a licence, alongside individuals, political parties and for-profit entities. A licensed station would otherwise have suited the requirement well — 100 W ERP, 5–10

km coverage, ₹2,500 application fee — and community radio may in any case only carry news sourced from Akashvani. **Implication:** FM broadcast is unavailable to the institution in its own name. A partner NGO holding a licence would not solve it either, since the content is religious and the station is not the institution's to program.

### 3. Spectrum — no licence-exempt band reaches 2 km with audio

India's [de-licensed bands](#) are 2.4 GHz, the 5 GHz sub-bands, lower 6 GHz, 865–867 MHz and 433–434 MHz, each at low power and each subject to its gazette notification. The 87.5–108 MHz entry on the short-range-device exemption list covers only very low power devices such as in-car FM adaptors, at field strengths measured at a few metres.

| Band            | Status                                                  | Why it fails here                                                                                                 |
|-----------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 87.5–108 MHz FM | Licensed for broadcast; SRD exemption is nanowatt-class | Institution ineligible to be licensed; exempt power covers metres, not 2 km                                       |
| 865–867 MHz     | Licence-exempt, low power                               | A few hundred kbit/s at best over short range; cannot carry continuous live audio to 500 nodes                    |
| 433–434 MHz     | Licence-exempt, very low power                          | Narrowband telemetry only                                                                                         |
| 2.4 / 5 / 6 GHz | Licence-exempt, low power                               | Realistic indoor reach from a tower is a few hundred metres, not 2 km, with 500 scattered non-line-of-sight homes |

**Implication:** there is no lawful, licence-exempt, institution-owned RF path to 500 homes at 2 km. This is the single finding that decides the architecture.

### 4. Telecom — voice and messaging are usable, with conditions

Outbound voice and SMS in India fall under TRAI's commercial communications regulations and the DLT framework. A sender needs [DLT registration](#) (about ₹5,900 one-time, ₹590 per CLI header, ₹1,180–2,360 annual renewal), templates and headers must be pre-registered, and calls to DND-registered numbers are scrubbed unless explicit consent is on record. **Implication:** every household must give recorded, revocable consent at enrolment, and the institution (or its platform partner) must hold DLT registration before the first voice broadcast. Consent also satisfies the Digital Personal Data Protection Act, 2023, which governs the household roster of names and phone numbers.

Nothing in this project requires a broadcasting licence: distributing audio over the public internet or over telephone calls to a consenting, enrolled membership is not broadcasting.

## Business objectives and success criteria

The project succeeds if the weekly message keeps arriving, at a cost the community can carry, without depending on a permission that can be withdrawn.

| #   | Objective                                        | Measure                                                    | Target at end of pilot (month 6) |
|-----|--------------------------------------------------|------------------------------------------------------------|----------------------------------|
| O1  | Continuity independent of loudspeaker permission | Weekly broadcasts delivered without a PA system            | 26 of 26 consecutive weeks       |
| O2  | Reach every enrolled household                   | Households receiving the live audio, any tier              | ≥ 90% of 500 enrolled            |
| O3  | Stay inside the household cost ceiling           | Highest one-time cost borne by any household               | ≤ ₹300, actual ₹285              |
| O4  | Stay inside the institution's capital envelope   | Delivered capital cost                                     | ₹1–4 lakh; Basic ≈ ₹1.42 lakh    |
| O5  | Keep the operator's job as simple as today       | Time from operator arriving to being live                  | ≤ 60 seconds, one button         |
| O6  | Work for households with no smartphone           | Non-smartphone households reached                          | ≥ 90% of that segment            |
| O7  | Sustainable running cost                         | Annual operating cost per household                        | ≤ ₹250                           |
| O8  | Prove delivery for the first time                | Broadcasts with a per-household delivery report            | 100%                             |
| O9  | Be ready to replicate                            | Second institution onboarded from documentation alone      | 1 by month 9                     |
| O10 | Run recurring daily broadcasts unattended        | Scheduled slots that open on time with no operator present | ≥ 99% of slots                   |
| O11 | Keep daily prayer from becoming intrusive        | Households that disable all slots after enrolling          | ≤ 5%                             |

**What is explicitly not an objective.** Replacing the loudspeaker while it is still permitted, broadcasting to the general public, generating revenue, or providing two-way communication. The channel is one-to-many, members-only, and additive.

**Failure conditions.** The project should be stopped or redesigned if, at the month-3 gate, fewer than 60% of enrolled households receive a given week's broadcast, or if the per-broadcast operator effort exceeds five minutes, or if the household device cost cannot be held under ₹300 at pilot volume.

## Stakeholders and personas

The design is constrained more by the least-equipped household than by the average one, so the personas are ordered by how much they limit the solution.

| Stakeholder                                       | Interest                                                       | Decides / constrains                     |
|---------------------------------------------------|----------------------------------------------------------------|------------------------------------------|
| Institution head (parish priest, imam, mutawalli) | Message reaches the flock; dignity of delivery; no controversy | Approves budget, content, tone           |
| Managing committee / trust                        | Capital and annual cost, accountability                        | Approves ₹1–4 lakh spend                 |
| Broadcast operator (volunteer, often 50+)         | Can start the broadcast without help                           | Hard cap on UI complexity                |
| Household members (500)                           | Hear the message at home, no cost, no fuss                     | Hard cap on device cost and setup effort |
| Neighbours of other faiths                        | Not to be subjected to amplified sound                         | Reason the project exists                |
| District authority / police                       | Compliance with noise rules                                    | Can withdraw PA permission               |
| Telecom / DLT regulator                           | Consent and header compliance for voice                        | Gates the T3 voice channel               |
| Platform operator (future)                        | Replicable, supportable across institutions                    | Shapes multi-tenant design               |

## Household personas

**P1 — Connected family (≈ 55%, 275 houses).** At least one adult with a smartphone and either home Wi-Fi or a data pack. Comfortable with WhatsApp; less so with app stores. Wants a link that just plays. **Endpoint:** T1, browser link or app, ₹0.

**P2 — Wi-Fi home, no willing smartphone user (≈ 30%, 150 houses).** Broadband exists, often for a child's study or a TV stick, but the person who listens is not the person who owns the phone. Needs an appliance that behaves like a radio: always on, no screen, one

knob. **Endpoint:** T2 Parish Box, ₹285.

**P3 — Feature phone or no home internet ( $\approx 15\%$ , 75 houses).** Often elderly, often living alone, often the households that rely on the loudspeaker most. Cannot install anything, cannot be trained on menus, but can answer a ringing phone. **Endpoint:** T3, the phone rings and the live audio plays; press nothing. ₹0.

**P4 — Members outside the 2 km radius.** Students, migrant workers, family abroad. Not reachable by the loudspeaker today, so they are pure upside. **Endpoint:** T1, at no extra cost.

P3 is the persona that eliminates most designs. Any solution requiring purchase, installation, charging, pairing or menu navigation fails for P3, and P3 is precisely the group the current loudspeaker serves best.

## Scope

### In scope

- One institution, up to 500 enrolled households, within roughly 2 km.
- Live one-way audio from a single origination point inside the institution, both on demand and on a recurring daily schedule of up to five adhan or their equivalent.
- Three household endpoints: browser/app (T1), Parish Box appliance (T2), telephone (T3).
- Household roster, consent capture, enrolment and opt-out.
- Operator console: one-button start and stop, live status, delivery report.
- Prayer or service timetable, per-household slot selection, per-slot volume and quiet windows.
- Documentation sufficient for a second institution to be onboarded without the original build team.

### Out of scope

- Two-way audio, call-in, Q&A or conferencing.
- Video of any kind.
- Replacing or modifying the existing loudspeaker system. It stays as it is.
- Public or open access. The channel is for enrolled members only.
- Emergency or disaster alerting as a certified service. The channel may carry urgent notices but is not a life-safety system and must not be described as one.
- Payments, donations, attendance or membership management.
- Content creation. The institution writes and speaks its own message.

- Multi-language automatic dubbing in the pilot. Transcription and text translation only, and only in Pro.

## Assumptions

| #  | Assumption                                                                                          | If wrong                                                                                           |
|----|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| A1 | About 85% of households have a smartphone or home broadband in the family                           | T3 voice volume rises; opex rises roughly ₹250 per extra household per year                        |
| A2 | Mains power is available at most homes for 6+ hours a day                                           | T2 boxes need a battery variant, adding ≈ ₹60 to BOM                                               |
| A3 | Mobile coverage is adequate for a voice call at every address                                       | Affected households need T2 with a neighbour's Wi-Fi, or a listening point                         |
| A4 | The institution can nominate two volunteers as operators                                            | Paid part-time operator, ≈ ₹36,000/year                                                            |
| A5 | Households will give consent for their phone number to be used                                      | Reach falls; consent drive must be led by the institution head, not a vendor                       |
| A6 | Load is one weekly announcement of 5–20 minutes plus up to five daily adhan of about 3 minutes each | Streaming and voice cost scale with total minutes; extra slots such as taraweeh add proportionally |
| A7 | Box BOM holds at ₹285 for a 500-unit order                                                          | Subsidy from the institution's capital budget covers the gap                                       |

## Dependencies

- A fixed broadband connection at the institution, with a 4G/5G router as failover.
- DLT telemarketer registration and a registered CLI header before any T3 voice broadcast.
- A cloud streaming host and a voice/IVR platform account.
- A hardware assembler willing to build 150–400 boxes to a fixed BOM.
- The institution's own household register as the enrolment starting point.

## Options evaluated

Eight transports were assessed against the five binding constraints. Only three survive, and the recommendation combines them.

| Option                                       | Legal for this institution                                    | Live                                            | Household one-time cost  | Reaches P3 (feature phone / elderly) | Verdict                                                                         |
|----------------------------------------------|---------------------------------------------------------------|-------------------------------------------------|--------------------------|--------------------------------------|---------------------------------------------------------------------------------|
| Licensed community FM, 100 W                 | <b>No</b> — religious bodies ineligible under CRS clause 2(b) | Yes                                             | ₹250 radio               | Yes                                  | <b>Rejected</b> on eligibility                                                  |
| Unlicensed FM transmitter                    | <b>No</b> — unlawful above nanowatt SRD limits                | Yes                                             | ₹250 radio               | Yes                                  | <b>Rejected;</b> criminal exposure                                              |
| 865–867 MHz / LoRa-class RF                  | Yes                                                           | No — bandwidth cannot carry speech to 500 nodes | ₹400+                    | Yes                                  | <b>Rejected</b> on physics                                                      |
| Institution-owned Wi-Fi mesh to homes        | Yes                                                           | Yes                                             | ₹285                     | Yes                                  | <b>Rejected</b> — realistic reach is a few hundred metres, not 2 km             |
| Cable TV audio channel via local operator    | Yes                                                           | Yes                                             | ₹0 if already subscribed | Partly                               | <b>Deferred</b> — viable where a single operator serves the area; check locally |
| Internet stream to own smartphone            | Yes                                                           | Yes                                             | ₹0                       | No                                   | <b>Selected</b> as T1                                                           |
| Internet stream to a dedicated Wi-Fi speaker | Yes                                                           | Yes                                             | ₹285                     | Yes, if home has Wi-Fi               | <b>Selected</b> as T2                                                           |
| Outbound phone call into a live bridge       | Yes, with consent + DLT                                       | Yes                                             | ₹0                       | Yes                                  | <b>Selected</b> as T3, for the residual segment                                 |

## Why no single option wins

Each surviving option fails a different constraint on its own. T1 is free but misses every household without a smartphone. T2 covers the appliance-preferring household but needs home Wi-Fi, which the poorest and oldest households are least likely to have. T3 needs nothing at all but costs money every minute of every broadcast, so it cannot carry all 500 households.

A 20-minute weekly broadcast to all 500 households by outbound call would cost roughly **₹2,400 a week**, or **₹1.25 lakh a year**, at ₹0.12 per 30-second pulse before GST. Restricting T3 to the residual 15% brings that to about **₹19,000 a year**, which the institution can sustain. Restricting T3 is therefore a cost decision, not a preference.

## Why replay is a Pro feature

Live-only is genuinely cheaper: no recording store, no archive hosting, no on-demand voice minutes, no app, no rights or retention policy, and no transcript workflow. It also matches the loudspeaker's behaviour exactly, which makes it easier to explain. Replay roughly doubles the capital cost and adds about ₹46,000 a year, which is why it belongs in Pro.

## What was not considered

DAB/DRM digital radio (no consumer receivers in India at this price), satellite radio (licensing and cost), and door-to-door printed notices (does not carry the voice, which is the point).

## Recommended architecture

One microphone, one encoder, one stream, three ways to hear it. The existing PA amplifier becomes the audio source, so nothing at the origination end is bought twice.

```
flowchart LR
    MIC[Existing mic<br/>and PA amp] --> ENC[Encoder<br/>at institution]
    BTN[One-button<br/>console] --> ENC
    ENC --> CLOUD[Stream relay<br/>in cloud]
    CLOUD --> T1[T1 phone<br/>app or link]
    CLOUD --> T2[T2 Parish Box<br/>Wi-Fi speaker]
    CLOUD --> GW[Voice gateway]
    GW --> T3[T3 any phone<br/>rings and plays]
    ENC --> PA[Loudspeakers<br/>unchanged]
```

## Origination (inside the institution)

The PA mixer's line output feeds a USB audio interface on a small encoder appliance. The encoder is a fanless single-board computer running an Opus/AAC encoder at 48 kbit/s mono, which is ample for speech and cheap to distribute. A wall-mounted console — a tablet in a locked kiosk app, or a physical illuminated button — starts and stops the broadcast. The operator does nothing else. A UPS holds the encoder, router and console through a power cut. A 4G router takes over automatically if the fixed line drops.

## Transport

The encoder pushes to a small cloud relay (Icecast or low-latency HLS behind a CDN). Target glass-to-ear latency is **≤ 8 seconds** for T1 and T2 and **≤ 4 seconds** for T3. Bandwidth is modest: 48 kbit/s × 500 listeners × 30 minutes weekly is about **300 GB a year**, which a ₹750/month VPS absorbs. The stream URL is unlisted and token-gated so it is a members' channel, not a public broadcast.

## Household endpoints

**T1 — phone.** A progressive web app opened from a saved link or a WhatsApp message. One large play button, the institution's name, and nothing else. In Pro, a native Android app adds a push notification and offline replay. No account; a one-time household token is embedded in the link.

**T2 — Parish Box.** A palm-sized mains-powered speaker with an ESP32-class Wi-Fi module, a volume knob, and a single status light. It is configured once at enrolment by a volunteer, then never touched. When a broadcast starts it wakes, connects, plays, and returns to standby. Green light means ready; amber means it cannot reach Wi-Fi. There is no screen, no app, no pairing and no firmware the household must manage.

**T3 — telephone.** At broadcast start, the voice gateway dials the enrolled numbers in this tier and bridges answered calls into the live audio. The household answers and listens. Unanswered numbers are retried twice. In Pro, the same number can be called back at any time during the week to hear the last message, which also gives every other tier a fallback.

## One broadcast, end to end

1. Operator arrives, presses the console button. The system announces itself as live within 10 seconds.
2. T1 and T2 devices are woken by a push and a wake packet; T2 boxes light green and open the stream.
3. The voice gateway begins dialling the T3 list, ramping to avoid carrier rate limits.
4. Operator speaks into the same microphone used for the loudspeakers.
5. Console shows a live count: devices connected, calls answered, calls failing.
6. Operator presses stop. Calls are released, boxes return to standby.
7. A delivery report is written: per household, tier, connected or not, duration.

A scheduled prayer slot skips steps 1 and 6: the system arms itself at the timetable time and closes itself after 30 seconds of silence, so nobody presses anything five times a day.

## Basic versus Pro

| Capability                                      | Basic             | Pro            |
|-------------------------------------------------|-------------------|----------------|
| Live stream to T1, T2, T3                       | Yes               | Yes            |
| One-button operator console                     | Yes               | Yes            |
| Delivery report                                 | Yes               | Yes            |
| Recording and archive                           | No                | Yes, 12 months |
| Replay on demand (app and dial-in)              | No                | Yes            |
| Native app with push                            | No                | Yes            |
| Transcript and text translation                 | No                | Yes            |
| Redundant encoder and auto-failover             | No                | Yes            |
| Analytics over time                             | Basic counts only | Yes            |
| Scheduled recurring broadcasts (adhan, Angelus) | Yes               | Yes            |

## Recurring daily broadcasts

The channel must carry up to **five adhan a day, every day**, not only a weekly message. This is a change in kind, not in degree: 35 broadcasts a week instead of one, several of them outside normal waking hours. Three consequences follow, and they reshape parts of this specification.

**1. Scheduling becomes mandatory, not a Pro feature.** Nobody presses a button five times a day. The system holds a prayer timetable and arms itself; the muezzin simply calls the adhan into the same microphone he uses now.

**2. The phone-call tier cannot carry adhan.** TRAI's time-band restriction on commercial communication effectively bars automated outbound calls before 9 am and after 9 pm, which rules out Fajr and Isha regardless of consent. Cost rules out the rest: five calls a day to 500 households would run to roughly **₹6.5 lakh a year** against ₹19,000 for a weekly-only call tier. **T3 households receive weekly announcements only.** Adhan reaches them only if they move to a Parish Box.

**3. Privacy indoors is the whole advantage.** A loudspeaker at 4:45 am is the exact use that noise rules and neighbour complaints target. The same adhan played on a speaker inside a consenting household's own home, at a volume that household chose, raises none of those issues. Daily prayer is therefore a stronger case for this channel than the weekly announcement is.

## How a scheduled broadcast runs

```
stateDiagram-v2
    [*] --> Standby
    Standby --> Armed: scheduled time
    Armed --> Live: muezzin speaks
    Armed --> Fallback: silence past grace
    Live --> Standby: silence 30s or 6min cap
    Fallback --> Standby: recording ends
```

The box wakes, plays a short chime so the household knows the channel is open, then plays the live call. If the microphone stays silent past a grace period, the system either falls back to a stored recording of the masjid's own muezzin or stays silent — configurable, and **off by default**, because whether a recorded adhan is acceptable is a matter for the institution, not the vendor.

## Timetable

| Requirement     | Detail                                                                                                                                                  |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source of times | Computed from the institution's coordinates, with the calculation method and madhhab set by the institution; or imported from its own printed timetable |
| Override        | Any single slot editable for that day, and any slot permanently offset from the computed time                                                           |
| Extra slots     | Iqamah, Jumu'ah khutbah, taraweeh, sehri and iftar in Ramadan; Angelus and daily mass for a church                                                      |
| Drift           | Times recomputed nightly; timetable published to households a month ahead                                                                               |
| Failure         | If the timetable service is unreachable, the last known schedule continues to run locally                                                               |

## Per-household control

Five daily broadcasts make household-level control essential — without it, households switch the box off and are lost.

- Each household chooses which of the five prayers it receives, and which other events.
- Per-slot volume, with Fajr defaulting to a lower level than the rest.

- A quiet window that suppresses everything except a notice the institution explicitly flags as urgent.
- All of this is settable by the office on the household's behalf, because the households that most need adhan are the least able to configure anything themselves.
- Households within earshot of the outdoor loudspeaker default to announcements only, so they do not hear the adhan twice a few seconds apart. Where both are wanted, a per-household delay offset of 0–15 seconds aligns the box with the outdoor sound.

## What this costs

|                                             | Weekly only | With five daily adhan               |
|---------------------------------------------|-------------|-------------------------------------|
| Broadcast minutes per week                  | 20          | ≈ 155                               |
| Household data per month (T1/T2, 32 kbit/s) | ≈ 20 MB     | ≈ 160 MB                            |
| Streaming bandwidth per year, 500 listeners | ≈ 300 GB    | ≈ 1 TB                              |
| Cloud hosting per year                      | ₹9,000      | ₹15,000                             |
| T3 voice per year                           | ₹19,000     | ₹19,000 — unchanged, adhan excluded |
| Operator time per week                      | 10 min      | 10 min — adhan slots are unattended |

The capital cost does not change. Operating cost rises by about **₹6,000 a year**, which is the cheapest part of this entire decision. The real cost of daily prayer is that the tier mix shifts: households that want adhan need a box or a smartphone, so the T3 share should be driven down before launch rather than after.

## Functional requirements

Priority: **M** must have for Basic, **S** should have for Basic, **P** Pro only.

## Capture and origination

| ID    | Requirement                                                                                | Pri |
|-------|--------------------------------------------------------------------------------------------|-----|
| FR-01 | Accept a line-level audio input from the existing PA mixer without modifying the PA system | M   |

| ID    | Requirement                                                                                                        | Pri |
|-------|--------------------------------------------------------------------------------------------------------------------|-----|
| FR-02 | Encode speech at 32 kbit/s mono by default, with a 48 kbit/s option, using automatic gain control and a noise gate | M   |
| FR-03 | Start and stop a broadcast from a single physical or on-screen control                                             | M   |
| FR-04 | Insert a recorded opening and closing chime so listeners know when the message begins and ends                     | M   |
| FR-05 | Mute the outbound stream on operator command without stopping the session                                          | S   |
| FR-06 | Accept a pre-recorded audio file as the source in place of the live microphone                                     | S   |

## Scheduled and recurring broadcasts

| ID    | Requirement                                                                                                                        | Pri |
|-------|------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-70 | Open a broadcast automatically at a scheduled time with no operator present                                                        | M   |
| FR-71 | Hold a prayer or service timetable, computed from the institution's coordinates with a configurable calculation method and madhhab | M   |
| FR-72 | Accept a manually entered or imported timetable that overrides the computed one                                                    | M   |
| FR-73 | Allow a permanent per-slot offset, and a one-off change to a single day's slot                                                     | M   |
| FR-74 | Support at least 8 slots a day, including iqamah, Jumu'ah, taraweeh, sehri and iftar                                               | M   |
| FR-75 | Recompute times nightly and continue on the last known schedule if the timetable service is unreachable                            | M   |
| FR-76 | Close the broadcast automatically after 30 s of silence, or at a hard cap of 6 minutes, whichever comes first                      | M   |

| ID    | Requirement                                                                                                | Pri |
|-------|------------------------------------------------------------------------------------------------------------|-----|
| FR-77 | Optionally play a stored recording if the microphone stays silent past a grace period, disabled by default | S   |
| FR-78 | Publish the coming month's schedule to households                                                          | S   |
| FR-79 | Suppress a scheduled slot for a single day on one operator action                                          | S   |

## Broadcast control and monitoring

| ID    | Requirement                                                                           | Pri |
|-------|---------------------------------------------------------------------------------------|-----|
| FR-10 | Show live status: on air or off air, in large type, readable across a room            | M   |
| FR-11 | Show live connected counts by tier, updating at least every 10 seconds                | M   |
| FR-12 | Warn the operator, before going live, if the encoder, network or gateway is unhealthy | M   |
| FR-13 | Continue the broadcast uninterrupted across a fixed-line-to-4G failover               | M   |
| FR-14 | Cap broadcast duration at a configurable limit and warn the operator before cut-off   | S   |
| FR-15 | Allow an authorised second operator to take over from a different device              | P   |

## Household reception

| ID    | Requirement                                                                    | Pri |
|-------|--------------------------------------------------------------------------------|-----|
| FR-20 | T1: play the live stream from a saved link with a single tap, no login         | M   |
| FR-21 | T1: notify the household when a broadcast begins, by push or WhatsApp          | M   |
| FR-22 | T2: connect and begin playing within 20 seconds of broadcast start, unattended | M   |

| ID    | Requirement                                                                                                      | Pri |
|-------|------------------------------------------------------------------------------------------------------------------|-----|
| FR-23 | T2: expose exactly one control (volume) and one indicator (status light)                                         | M   |
| FR-24 | T2: recover automatically from power loss and Wi-Fi loss with no household action                                | M   |
| FR-25 | T2: be configurable for a new Wi-Fi password by a volunteer in under 3 minutes                                   | M   |
| FR-26 | T3: dial enrolled numbers at broadcast start and bridge answered calls to live audio                             | M   |
| FR-27 | T3: retry unanswered numbers twice, at 3 and 8 minutes                                                           | M   |
| FR-28 | T3: place outbound calls only between 9 am and 9 pm, and never for a scheduled prayer slot                       | M   |
| FR-29 | Any tier: a household may opt out, change tier, or change which slots it receives with one request to the office | M   |
| FR-30 | T2: fall back to a household's mobile hotspot if configured                                                      | S   |

## Per-household slot control

Without these, households facing five broadcasts a day switch the box off and are lost from the channel entirely.

| ID    | Requirement                                                                                               | Pri |
|-------|-----------------------------------------------------------------------------------------------------------|-----|
| FR-31 | Let a household choose which scheduled slots it receives, independently of the weekly announcement        | M   |
| FR-32 | Hold a per-slot volume level, with Fajr defaulting lower than the other prayers                           | M   |
| FR-33 | Honour a household quiet window that suppresses all slots except a notice the institution flags as urgent | M   |
| FR-34 | Apply a per-household playback delay of 0–15 s, so a box can be aligned with the outdoor loudspeaker      | S   |

| ID    | Requirement                                                                                          | Pri |
|-------|------------------------------------------------------------------------------------------------------|-----|
| FR-35 | Let the office set all of the above on a household's behalf, without the household touching anything | M   |
| FR-36 | Default households within earshot of the outdoor loudspeaker to announcements only                   | S   |

## Replay and archive (Pro)

| ID    | Requirement                                                        | Pri |
|-------|--------------------------------------------------------------------|-----|
| FR-40 | Record every broadcast automatically and retain for 12 months      | P   |
| FR-41 | Make the last broadcast playable on demand in the app              | P   |
| FR-42 | Let any phone dial a number and hear the most recent broadcast     | P   |
| FR-43 | Publish an archive list by date and title, searchable              | P   |
| FR-44 | Generate a text transcript in the local language within 30 minutes | P   |
| FR-45 | Offer the transcript translated into one additional language       | P   |
| FR-46 | Allow the institution to delete any recording permanently          | P   |

## Roster, enrolment and consent

| ID    | Requirement                                                                                  | Pri |
|-------|----------------------------------------------------------------------------------------------|-----|
| FR-50 | Maintain a household register: household name, address, tier, contact numbers, status        | M   |
| FR-51 | Record explicit, dated consent per household for voice contact, with the consent text stored | M   |
| FR-52 | Support bulk import from the institution's existing register                                 | M   |

| ID    | Requirement                                                                       | Pri |
|-------|-----------------------------------------------------------------------------------|-----|
| FR-53 | Assign and reassign a household's tier without re-enrolment                       | M   |
| FR-54 | Record and honour opt-out immediately, and suppress that household from all tiers | M   |
| FR-55 | Bind each T2 box to exactly one household record, by device identifier            | M   |
| FR-56 | Support households with more than one listening endpoint                          | S   |

## Administration and reporting

| ID    | Requirement                                                                                | Pri |
|-------|--------------------------------------------------------------------------------------------|-----|
| FR-60 | Produce a per-broadcast delivery report: household, tier, connected, duration              | M   |
| FR-61 | Flag households that missed three consecutive broadcasts, for a volunteer visit            | M   |
| FR-62 | Show running cost for the month against budget                                             | S   |
| FR-63 | Maintain an audit log of who broadcast, when, and for how long                             | M   |
| FR-64 | Operate in the institution's own language throughout the operator and household interfaces | M   |
| FR-65 | Support more than one institution on the same platform, with strict data separation        | P   |

## Non-functional requirements

| ID     | Area    | Requirement                                                                                                            |
|--------|---------|------------------------------------------------------------------------------------------------------------------------|
| NFR-01 | Latency | Glass-to-ear $\leq 8$ s for T1 and T2 on announcements, $\leq 4$ s on scheduled adhan, measured at the 95th percentile |

| ID     | Area                  | Requirement                                                                                                                                           |
|--------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-02 | Latency spread        | All endpoints within 3 s of each other, so neighbours do not hear an echo across a street — tighter than for announcements because adhan recurs daily |
| NFR-03 | Availability          | 99% of scheduled weekly broadcasts start within 2 minutes of the intended time                                                                        |
| NFR-04 | Audio quality         | Speech intelligible at 48 kbit/s mono; no dropout longer than 2 s more than once per broadcast                                                        |
| NFR-05 | Capacity              | 500 concurrent listeners per institution, with headroom to 750                                                                                        |
| NFR-06 | Voice ramp            | T3 dialling completes to 95% of the list within 4 minutes of broadcast start                                                                          |
| NFR-07 | Power resilience      | Origination survives a 90-minute mains outage on UPS; T2 resumes automatically when power returns                                                     |
| NFR-08 | Network resilience    | Fixed-to-mobile failover completes within 15 s with no operator action                                                                                |
| NFR-09 | Degradation           | If the stream fails, the system falls back to voice calls for all tiers rather than failing silent                                                    |
| NFR-10 | Usability — operator  | A volunteer aged 60+ can go live unaided after one 10-minute training session                                                                         |
| NFR-11 | Usability — household | T2 requires zero household action after installation; T3 requires only answering a call                                                               |
| NFR-12 | Accessibility         | Status conveyed by both colour and shape or sound; volume range adequate for hearing loss                                                             |
| NFR-13 | Language              | All household-facing and operator-facing text in the institution's local language                                                                     |
| NFR-14 | Privacy               | Household names and numbers held under DPDP Act 2023: explicit consent, stated purpose, deletion on request                                           |
| NFR-15 | Data minimisation     | No location tracking, no listening history beyond connected or not, no contact-list access                                                            |

| ID     | Area              | Requirement                                                                                                    |
|--------|-------------------|----------------------------------------------------------------------------------------------------------------|
| NFR-16 | Security          | Stream URLs token-gated and rotated; console access password-protected; no default credentials on any device   |
| NFR-17 | Content integrity | Only the institution's own console can originate; no remote injection path into the stream                     |
| NFR-18 | Maintainability   | T2 firmware updates over the air, staged, with automatic rollback on failure                                   |
| NFR-19 | Supportability    | A failed box is swapped, not repaired; the institution holds 5% spares                                         |
| NFR-20 | Cost control      | A hard monthly cap on voice spend, with the operator warned before it is reached                               |
| NFR-21 | Scalability       | Platform supports 50 institutions without architectural change; per-institution data isolated                  |
| NFR-22 | Portability       | No dependence on a single telecom or cloud vendor; stream and voice providers replaceable in under a week      |
| NFR-23 | Compliance        | Operates without any broadcasting licence; DLT registration and DND consent verified before each voice run     |
| NFR-24 | Environmental     | T2 draws under 0.5 W in standby and under 2 W playing; a box costs about ₹30 a year to run at five slots a day |

## Household cost model

The binding number is ₹300 one-time. Two of the three tiers cost the household nothing at all; only the Parish Box has a bill of materials, and it lands at **₹285** at a 500-unit order.

### T2 Parish Box — bill of materials, 500-unit order

| Item             | Spec                                   | ₹  |
|------------------|----------------------------------------|----|
| Wi-Fi SoC module | ESP32-C3 class, 2.4 GHz, I2S audio out | 95 |
| Audio amplifier  | 3 W class-D (PAM8403 class)            | 14 |

| Item                                            | Spec                             | ₹          |
|-------------------------------------------------|----------------------------------|------------|
| Speaker                                         | 66 mm, 3 W, 4 Ω, full-range      | 32         |
| Enclosure                                       | ABS, wall or shelf mount, vented | 42         |
| Power supply                                    | 5 V 1 A adaptor, BIS-marked      | 52         |
| PCB, volume pot, status LED, connectors, wiring |                                  | 28         |
| Assembly, flashing and functional test          | Contract assembly at 500 units   | 22         |
| <b>Total</b>                                    |                                  | <b>285</b> |

Margin to the ceiling is ₹15, which is thin. Two levers protect it: ordering 1,000 units across two institutions cuts the module and enclosure lines by roughly 12%, and the institution's capital budget can absorb any overrun so the household price stays fixed at ₹300 regardless of BOM drift. **The household price is a promise; the BOM is an estimate.**

### Cost by tier

| Tier          | Households | One-time to household | Recurring to household                              | Who pays the rest             |
|---------------|------------|-----------------------|-----------------------------------------------------|-------------------------------|
| T1 phone      | 275        | ₹0                    | Own data, 20–160 MB/month depending on slots chosen | Institution pays streaming    |
| T2 box        | 150        | ₹285                  | Own Wi-Fi (already paid) + ≈ ₹30/year electricity   | Institution pays streaming    |
| T3 phone call | 75         | ₹0                    | ₹0 — incoming calls are free in India               | Institution pays call charges |

### Data use, T1 and T2

At 32 kbit/s, a 20-minute broadcast is about **5 MB** and a 3-minute adhan about **0.7 MB**. A household on the weekly announcement alone uses roughly **20 MB a month**. A household taking all five daily adhan as well uses roughly **160 MB a month** — still inside any ordinary prepaid pack, but worth stating plainly at enrolment, because a perception that “it eats data” will kill adoption faster than any technical fault. Households on a metered connection can be set to fewer slots.

## Subsidy policy

The committee should decide one of three positions before enrolment begins, since it changes the enrolment script:

1. **Full price, ₹300 per box.** Maximum cost recovery, likely lowest uptake among the households that need it most.
2. **Sliding scale.** ₹300 default, waived on request by the institution head, no questions asked. Recommended.
3. **Free to all.** Simplest to explain; adds about ₹42,750 to the capital cost for 150 boxes, which the ₹1–4 lakh envelope can absorb.

The cost model in this document assumes option 3 for budgeting, so the capital figures are the worst case for the institution and the best case for the household.

## Institution cost model

Basic lands at **₹1.46 lakh** and Pro at **₹3.48 lakh**, both inside the ₹1–4 lakh envelope. The largest single line in Basic is the boxes, not the electronics, which is the right place for the money to go.

### Capital — Basic (live only)

| Item                                                         | ₹      |
|--------------------------------------------------------------|--------|
| USB audio interface from existing PA mixer output            | 2,200  |
| Encoder appliance (SBC, case, PSU, storage)                  | 9,500  |
| Operator console tablet in locked kiosk mount                | 12,000 |
| 4G failover router (fixed line provided separately)          | 6,500  |
| UPS 600 VA with battery                                      | 6,000  |
| Rack, cabling, surge protection, earthing                    | 5,500  |
| Cloud streaming host, first year prepaid                     | 15,000 |
| Parish Boxes — 150 units at ₹285                             | 42,750 |
| Enrolment drive, printed guides, signage, volunteer training | 9,000  |
| Integration, commissioning and acceptance testing            | 18,000 |
| DLT telemarketer registration and CLI header                 | 6,500  |

| Item               | ₹                |
|--------------------|------------------|
| Contingency, 10%   | 13,300           |
| <b>Basic total</b> | <b>₹1,46,250</b> |

### Capital — Pro increment (adds replay, app, redundancy)

| Item                                             | ₹                |
|--------------------------------------------------|------------------|
| Redundant encoder with automatic failover        | 11,000           |
| Archive storage, 2 TB, on-site plus cloud copy   | 14,000           |
| Additional 250 Parish Boxes                      | 71,250           |
| Replay IVR platform setup                        | 12,000           |
| Native Android app build and store listing       | 65,000           |
| Transcription and translation engine, first year | 22,000           |
| Analytics dashboard                              | 18,000           |
| Solar panel and battery for the encoder rack     | 28,000           |
| Extended support contract, first year            | 24,000           |
| Contingency, 10%                                 | 26,500           |
| <b>Pro increment</b>                             | <b>₹2,01,750</b> |
| <b>Pro total (Basic + increment)</b>             | <b>₹3,48,000</b> |

### Annual operating cost

| Item                                                 | Basic ₹/yr | Pro ₹/yr |
|------------------------------------------------------|------------|----------|
| Cloud streaming host and bandwidth ( $\approx$ 1 TB) | 15,000     | 20,000   |
| Broadband and 4G data at the institution             | 14,400     | 14,400   |
| T3 voice minutes (75 households, 20 min weekly)      | 19,000     | 19,000   |
| Replay dial-in minutes                               | —          | 9,000    |
| Notifications (SMS / WhatsApp)                       | 6,000      | 6,000    |
| DLT annual renewal                                   | 2,400      | 2,400    |

| Item                           | Basic ₹/yr     | Pro ₹/yr         |
|--------------------------------|----------------|------------------|
| Box replacement at 5% of fleet | 7,200          | 7,200            |
| Transcription and translation  | —              | 22,000           |
| App maintenance and store fees | —              | 15,000           |
| Electricity and consumables    | 4,000          | 4,000            |
| Support retainer               | 12,000         | 12,000           |
| <b>Total</b>                   | <b>₹80,000</b> | <b>₹1,31,000</b> |

## Five-year total cost of ownership

|                        | Basic            | Pro               |
|------------------------|------------------|-------------------|
| Capital                | ₹1,46,250        | ₹3,48,000         |
| Operating, 5 years     | ₹4,00,000        | ₹6,55,000         |
| <b>Five-year total</b> | <b>₹5,46,250</b> | <b>₹10,03,000</b> |
| Per household per year | <b>₹219</b>      | <b>₹401</b>       |

At about ₹18 per household per month, Basic costs a household less than one cup of tea a month, and that now buys five daily adhan plus the weekly announcement. Pro is roughly double.

## Cost levers if the budget tightens

- Drop the console tablet for a ₹1,500 illuminated button and a phone-based status page: saves ₹10,500.
- Start with 80 boxes instead of 150 and add as demand proves itself: saves ₹20,000 capital.
- Cap T3 at 12 minutes per broadcast rather than 20: saves about ₹7,600 a year.
- Self-host the stream on the encoder behind a dynamic DNS instead of a cloud relay: saves ₹15,000 a year, at the cost of resilience. Not recommended for the pilot.

## Operating model

The weekly broadcast must cost the institution no more effort than switching on the amplifier does today, or it will quietly stop happening.

## The weekly cycle

flowchart TD

```

A[Content agreed<br/>2 days before] --> B[Reminder sent<br/>2 hours before]
B --> C[Operator presses<br/>Go Live]
C --> D[Broadcast<br/>5-20 min]
D --> E[Operator presses<br/>Stop]
E --> F[Delivery report<br/>auto-generated]
F --> G[Volunteer follows up<br/>3-week misses]
G --> A

```

The five daily prayer slots sit outside this cycle entirely. They open and close on the timetable with no human in the loop, and appear in the operator's view only when one of them fails.

## Roles

| Role               | Who                                            | Time per week                            |
|--------------------|------------------------------------------------|------------------------------------------|
| Content owner      | Institution head                               | Already spent — same message as today    |
| Broadcast operator | Two trained volunteers, one primary one backup | 10 minutes                               |
| Roster keeper      | Office clerk or secretary                      | 20 minutes                               |
| Field volunteer    | 2-3 people, one per neighbourhood cluster      | 30 minutes, only when misses are flagged |
| Technical support  | Vendor retainer, remote                        | On call                                  |

No full-time staff. No technician on site. If a broadcast requires a technician, the design has failed NFR-10.

## Enrolment and onboarding

Enrolment is a one-time campaign, best run over three or four weekends after services, where the households already are.

1. The institution head announces the channel from the pulpit or minbar, and explains why. This announcement, not the technology, is what determines uptake.
2. A desk after the service captures: household name, address cluster, contact numbers, preferred tier, and signed consent for voice contact.

3. Households choosing T1 have the link saved to their phone and are shown it playing before they leave.
4. Households choosing T2 book a home visit. A volunteer installs the box, joins it to the home Wi-Fi, sets the volume, and tests a broadcast before leaving.
5. Households choosing T3 receive a test call the same day, so the number is verified while someone is still there to correct it.
6. Any household not enrolled after the campaign is visited once. Households that decline are recorded as declined, not chased.

## Support

- One WhatsApp number and one office phone number, published on a sticker on every box.
- A failed box is swapped from the spares pool within a week, never repaired in the home.
- Three consecutive misses on the delivery report triggers a visit, which is how silent failures are caught. This is the single most important operational control in the design.

## Content governance

- Only the institution head or a person they name may originate a broadcast.
- The channel carries community and religious announcements only. No political content, no commercial content, no third-party appeals.
- Content that would be unlawful on a loudspeaker is unlawful here. The private channel is not a way around content norms, and the committee should say so in writing at the outset.
- Broadcasts stay within a published window, by convention the same hours the loudspeaker uses today, even though no rule requires it. Voice calls are hard-blocked between 10 pm and 6 am.
- A complaints route exists and is published: any household may object to content to the committee.

## Rollout plan

Five phases over nine months, with a stop-or-go gate at each boundary. The loudspeaker keeps running throughout, so no phase carries the risk of silence.

| Phase                    | Weeks | Households               | What is built                                                          | Exit gate                                                                                                                   |
|--------------------------|-------|--------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 0 — Prove the stream     | 1–4   | 10 staff and volunteers  | Encoder, cloud relay, T1 link, console button, timetable and scheduler | 4 consecutive live broadcasts and 7 days of scheduled slots opening on time, latency within target, no operator help needed |
| 1 — Prove the box        | 5–10  | 25                       | 25 Parish Boxes, home installs, OTA update path                        | 90% of boxes wake and play for every scheduled slot over 4 consecutive weeks; install time $\leq 20$ min                    |
| 2 — Prove the phone tier | 11–16 | 60 (25 box + 35 voice)   | DLT registration, voice gateway, consent capture, retry logic          | 85% of T3 numbers answered or retried successfully; zero DND violations                                                     |
| 3 — Scale to full pilot  | 17–26 | 500                      | Full enrolment campaign, 150 boxes, delivery reports, follow-up loop   | O2 met: $\geq 90\%$ of enrolled households reached, 26 weeks unbroken                                                       |
| 4 — Pro and replicate    | 27–36 | 500 + second institution | Replay, app, archive, transcripts; onboarding pack                     | Second institution live from documentation alone                                                                            |

## What must be true before Phase 3

Phase 3 is the expensive phase — it commits the box order and the enrolment campaign. It should not start unless all of these hold:

- ☐ Phases 0–2 gates all passed
- ☐ Box BOM confirmed at  $\leq ₹300$  by a written supplier quote for 150 units, and D7 settled — whether T3 households are funded onto boxes for daily prayer
- ☐ DLT registration and CLI header live and tested
- ☐ Two volunteer operators trained and each has run a live broadcast alone
- ☐ Committee has signed off the subsidy policy and the content governance rules
- ☐ Consent wording reviewed against the DPDP Act and adopted

## Timeline shape

The critical path runs through DLT registration and the hardware order, not the software. Both should start in Phase 1, well before they are needed, since DLT approval can take several weeks and a 150-unit assembly run takes four to six.

## Before scaling to a second institution

Replication is a documentation problem, not an engineering one. Before institution two, the following must exist and have been used by someone other than the author: an install runbook for the Parish Box, an enrolment day script, an operator one-page card, a supplier list with current prices, and a fault-triage sheet. If the original team has to attend institution two's first broadcast, the pack is not finished.

## Risks and mitigations

The two risks most likely to kill this project are not technical: households not enrolling, and the broadcast quietly stopping once the novelty wears off.

| #  | Risk                                                                                           | L      | I      | Mitigation                                                                                                                                    |
|----|------------------------------------------------------------------------------------------------|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| R1 | Households do not enrol — the channel is seen as unnecessary while the loudspeaker still works | High   | High   | Institution head leads the announcement personally; enrol at services, not by visit; frame as insurance, not replacement; make T1 and T3 free |
| R2 | Broadcasts lapse after the pilot's attention fades                                             | High   | High   | Two trained operators, not one; delivery report reviewed monthly by the committee; the same weekly slot every week without exception          |
| R3 | Voice cost overruns as T3 grows beyond 15% of households                                       | Medium | Medium | Hard monthly spend cap with operator warning; duration cap; actively migrate T3 households to T2 where Wi-Fi exists                           |
| R4 | Box BOM rises above ₹300 before the order is placed                                            | Medium | Low    | Household price fixed at ₹300 and any gap absorbed by capital budget; order at 1,000 units across two institutions                            |

| #   | Risk                                                                 | L      | I      | Mitigation                                                                                                                     |
|-----|----------------------------------------------------------------------|--------|--------|--------------------------------------------------------------------------------------------------------------------------------|
| R5  | A household's Wi-Fi password changes and the box silently stops      | High   | Medium | Three-miss follow-up rule; amber status light; volunteer reconfigures in under 3 minutes                                       |
| R6  | Power cut during the broadcast window                                | High   | Medium | UPS at origination; T2 auto-resumes; T3 unaffected; the message is repeated the following week if a cut exceeded the broadcast |
| R7  | DLT registration or DND scrubbing blocks the voice tier              | Medium | High   | Start registration in Phase 1; capture explicit consent at enrolment; verify a test list before Phase 2                        |
| R8  | Loudspeaker permission withdrawn before the pilot completes          | Medium | High   | Phases are ordered so a usable channel exists from week 10; accelerate enrolment if a restriction is signalled                 |
| R9  | The private channel is perceived as evading noise regulation         | Low    | High   | State publicly that it is additive and compliant; brief the local authority before Phase 3; keep content within the same hours |
| R10 | Content misuse — the channel used for political or divisive messages | Low    | High   | Written governance rules adopted before launch; single named originator; audit log of every broadcast                          |
| R11 | Household data leaked or misused                                     | Low    | High   | DPDP-compliant consent and retention; no listening history; roster access limited to the clerk and the head                    |
| R12 | Cloud or telecom vendor raises prices or withdraws                   | Medium | Medium | No proprietary lock-in; standard Icecast/HLS and standard SIP; provider replaceable in under a week                            |
| R13 | Latency spread causes an audible echo between neighbouring houses    | Medium | Low    | NFR-02 caps the spread at 5 s; test across tiers in Phase 2; T3 buffer tuned to match the stream                               |

| #   | Risk                                                                                               | L      | I      | Mitigation                                                                                                                                                   |
|-----|----------------------------------------------------------------------------------------------------|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R14 | Boxes stolen, sold or repurposed                                                                   | Low    | Low    | Device bound to one household record; low resale value; written off at 5% a year in the opex model                                                           |
| R15 | Volunteers unavailable during festivals, exactly when broadcasts matter most                       | Medium | Medium | Backup operator rostered; Pro adds scheduled pre-recorded broadcasts for known absences                                                                      |
| R16 | Five broadcasts a day is felt as intrusive and households switch the box off                       | High   | High   | Per-household slot selection and per-slot volume from day one (FR-31 to FR-36); Fajr quieter by default; the office configures it for households that cannot |
| R17 | The scheduled slot opens but the muezzin is absent, and 500 homes hear silence or background noise | Medium | Medium | Auto-close on 30 s of silence; optional stored-recording fallback, off unless the institution enables it; missed-slot alert to the operator                  |
| R18 | A timetable error puts adhan at the wrong time, repeatedly and publicly                            | Low    | High   | Times recomputed nightly and shown on the console a week ahead; any slot overridable; the first month checked against the masjid's printed timetable         |
| R19 | Recorded adhan is theologically unacceptable to part of the community                              | Medium | Medium | Fallback recording disabled by default and enabled only on the institution head's written instruction                                                        |
| R20 | Households near the institution hear the outdoor adhan and the box a few seconds apart             | High   | Low    | Those households default to announcements only; per-household delay offset of 0–15 s where both are wanted                                                   |

## The risk worth dwelling on

R1 and R2 compound. If enrolment is weak, the delivery report looks poor; if the report looks poor, the committee loses interest; if the committee loses interest, the broadcasts lapse — and the channel is dead exactly when the loudspeaker permission becomes a live problem. Both mitigations are social rather than technical, and both depend on the institution head's personal visible commitment. No amount of engineering substitutes for it.

## Open decisions

Six decisions block the build. The first three change the cost model; the rest change the plan.

| #  | Decision                                               | Options                                                                                                   | Needed by                 |
|----|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------|
| D1 | Subsidy policy for the Parish Box                      | ₹300 full price / sliding scale with waiver / free to all                                                 | Before enrolment, Phase 3 |
| D2 | Basic or Pro at launch                                 | Start Basic and upgrade at month 9, or build Pro from the outset                                          | Before Phase 0            |
| D3 | Who holds DLT registration                             | The institution itself, or a platform partner acting as sender                                            | Phase 1                   |
| D4 | Target institution and its actual device mix           | The 55 / 30 / 15 split is an assumption; a 100-household survey would replace it                          | Before Phase 3 budgeting  |
| D5 | Whether to brief the district authority before Phase 3 | Brief proactively, or proceed and respond if asked                                                        | Phase 2                   |
| D6 | Second institution for replication                     | Same denomination and area, or deliberately different, to test the onboarding pack                        | Phase 4                   |
| D7 | Whether T3 households get adhan at all                 | Leave them on weekly announcements only, or fund a Parish Box for every household that wants daily prayer | Before Phase 3            |

| #  | Decision                                   | Options                                                                  | Needed by        |
|----|--------------------------------------------|--------------------------------------------------------------------------|------------------|
| D8 | Fallback recording of adhan                | Off permanently, or enabled using the institution's own muezzin recorded | Before Phase 0   |
| D9 | Which slots are on by default at enrolment | All five prayers, or Fajr off by default and opted into                  | Before enrolment |

## Questions this document could not answer

- **Actual smartphone and broadband penetration** in the target community. Every cost figure depends on the tier split, and the split here is assumed. A short survey at one service would settle it and is the single highest-value next step.
- **Whether a local cable TV operator serves the area.** If one operator covers most of the 500 homes, an audio channel on their network could be cheaper than T2 and reach P3 households directly. Worth one phone call before Phase 1.
- **Current status of the institution's loudspeaker permission** — granted, pending, conditional, or contested. This sets the urgency of the whole project and should be established first.
- **Whether the community is one language or several.** Multiple languages would move transcription and translation from Pro into Basic.

## Sources

- [Revised Policy Guidelines for setting up Community Radio Stations in India, 13.02.2024 — Ministry of Information and Broadcasting](#) — eligibility, clause 2(b) exclusion of religious bodies, 100 W ERP, fees
- [Loudspeakers in religious places: what does the law say — BOOM](#) — Noise Pollution Rules 2000, permission and night restrictions, decibel limits, court rulings
- [WPC de-licensed frequency bands in India — PCN Global India](#) — licence-exempt bands and their gazette notifications
- [Voice SMS services pricing in India 2026 — LeadNXT](#) — per-pulse voice rates, DLT registration and CLI header fees

Hardware prices, cloud hosting and labour figures are planning estimates from prevailing Indian market rates at September 2026 and are not sourced from the pages above. They must be re-quoted before any purchase order.